

Project Report: Smart Hospital & Resource Allocation System

Course: CSC 1012 Introduction to Computer Programming

1. High-Level Architecture and Implemented Functions

The system is designed using a modular, menu-driven approach within a single `main.c` file and a header file `hospital.h`. Instead of using separate user-defined functions for each task, the program flow is controlled via a do-while loop and a switch-case statement in the main function to handle user choices.

- **Initialization:** Loads previous bed occupancy data from `beds_status.txt`.
- **Menu System:** Displays the main menu and captures user input.
- **Patient Registration (Case 1):** Captures details, checks capacity, allocates a bed if admitted, and appends the record to `patient_records.txt`.
- **Print Patient Bill (Case 2):** Calculates surcharges, ward costs, age discounts, and wait times based on given formulas.
- **View Priority List (Case 3):** Uses Bubble Sort to sort patients by urgency level.
- **View Analytics Reports (Case 4):** Iterates through patient data to calculate total revenue, bed occupancy percentages, and find the highest-paying patient.
- **Exit & Save (Case 5):** Saves current bed occupancy back to `beds_status.txt` before terminating.

```
=====
SMART HOSPITAL & RESOURCE ALLOCATION SYSTEM
=====

1. Register a New Patient
2. Print Patient Bill
3. View Priority List
4. View Analytics Reports
5. Exit

Enter your choice: |
```

- **Input Validation & Outlier Handling:** Implemented while loops across data entry sections (such as Patient Registration and Patient ID searching) to prevent invalid user inputs. If a user enters an out-of-range value (e.g., negative age, invalid specialty ID, or wrong urgency level), the system displays a clear error message in red and continuously prompts the user until a valid input is provided.

```
Enter your choice: 1
```

```
- Patient Registration -----
```

```
Enter Patient Name: AMARA
```

```
Enter Patient Age: 150
```

```
You Entered Wrong Age! Please enter a valid age (1-119): |
```

```
Enter your choice: 1
```

```
- Patient Registration -----
```

```
Enter Patient Name: AMARA
```

```
Enter Patient Age: 22
```

```
Enter Urgency Level (1 = Normal, 2 = Urgent, 3 = Critical) : 2
```

```
Select Specialty (1=General, 2=Paediatrics, 3=Cardiology, 4=Neurology): 3
```

```
Is Admitted to Ward? (1 = Yes, 0 = No): 1
```

```
Select Ward ID (1=General, 2=Paediatric, 3=Surgical, 4=ICU): 3
```

```
Enter Days Admitted: -5
```

```
You Entered Wrong #day count! Please enter a valid Number : 2
```

2. Data Structure Designs

Since C structs were optional for this assignment, I implemented the data management using parallel 1D arrays and 2D arrays.

- **Parallel 1D Arrays:** Used to store patient records. The index `i` acts as the unique identifier for a single patient across all arrays.
- `char patientNames[MAX_PATIENTS][50];`
- `int patientAges[MAX_PATIENTS], daysAdmitted[MAX_PATIENTS];`
- `short urgencyLevel[MAX_PATIENTS], specialtyID[MAX_PATIENTS], isAdmitted[MAX_PATIENTS], wardID[MAX_PATIENTS];`
- **2D Array:** Used for tracking bed availability.
- `int bedOccupancy[4][20];` (Rows represent the 4 wards, columns represent the beds).
- **Constant Arrays:** Stored in `hospital.h` for quick lookup of base fees, consultation times, and ward rates.

```
Enter your choice: 1

- Patient Registration -----

Enter Patient Name: AMARA
Enter Patient Age: 22
Enter Urgency Level (1 = Normal, 2 = Urgent, 3 = Critical) : 2
Select Specialty (1=General, 2=Paediatrics, 3=Cardiology, 4=Neurology): 3
Is Admitted to Ward? (1 = Yes, 0 = No): 1
Select Ward ID (1=General, 2=Paediatric, 3=Surgical, 4=ICU): 3
Enter Days Admitted: 2

Bed #01 allocated successfully in Surgical Ward.

~ Patient Registered Successfully !!
```

3. Assumptions Made During Development

- The system handles a maximum of 100 patients per run (MAX_PATIENTS is defined as 100).
- Input values for menus (like choosing 1 to 5) are assumed to be valid integers provided by the user.

```
=====
SMART HOSPITAL & RESOURCE ALLOCATION SYSTEM
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1. Register a New Patient
2. Print Patient Bill
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4. View Analytics Reports
5. Exit

Enter your choice: 6

Invalid Choice #NUMBER! Please try again.
```

- Queue calculation for waiting time assumes all previously registered patients for that specialty are still waiting.
- Highest Bill calculation in analytics simply takes the first maximum value found if there is a tie.

```
Enter your choice: 2

- Print Patient Bill -----

Enter Patient ID (maximum 100): 1

=====
SMART HOSPITAL ADMISSION & BILL
=====
Patient ID      : PATIENT - 1
Patient Name    : AMARA
Age             : 22 Years (No Subsidy)
Specialty       : Cardiology
Assigned Ward   : Surgical Ward
Urgency Level   : Level 2 (Urgent)
Base Consultation Fee: LKR 4500.00
Emergency Surcharge : LKR 900.00 (20%)
Ward Stay Cost  : LKR 24000.00 (2 Days)
Gross Total Bill : LKR 29400.00
Age Subsidy Discount : LKR 0.00
Final Payable Amount : LKR 29400.00
Estimated Wait Time : 0.00 mins (Immediate Attention)
=====
```

Enter your choice: 4

- Analytics Reports -----

1. Patients: 1 (Normal:0, Urgent:1, Critical:0)
2. Revenue: LKR 29400.00 | Discounts: LKR 0.00
3. Bed Occupancy:
 - General Ward: 5.0%
 - Paediatric Ward: 0.0%
 - Surgical Ward: 20.0%
 - ICU: 0.0%
4. Highest Bill: AMARA (LKR 29400.00)

4. GitHub Repository

<https://github.com/janithchamika/Smart-Hospital-System.git>